

# CATS<sup>+</sup>

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## Competishum Advance

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### Training Sessions

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(Lecture - IV)



**Paragraph** Let  $x$  is a complex number satisfying the equation  $x \cdot \left( \frac{5-x}{x+1} \right) \cdot \left( x + \frac{\bar{2}-x}{x+1} \right) = 6$ , now let  $p = \max. \{ |\alpha_i| \}, i=1,2,3,4$  and  $q = \sum |\alpha_i|$ , for  $i=1,2,3,4$  where  $\alpha_i$ 's are distinct possible values of  $x$  satisfying the given equation then answer the followings:-

**Que-1** Value of  $p$  is

☐ A 1

☒ B 2

☐ C 3

☐ D None of these

**Que-2** Value of  $q$  is

☐ A 1

☐ B 2

☐ C 13

☒ D None of these.



Soln.

$$x \left( \frac{5-x}{x+1} \right) \left( x + \frac{5-x}{x+1} \right) = 6$$

$$(5x - x^2)(x^2 + 5) = 6(x^2 + 2x + 1)$$

$$x^4 - 5x^3 + 11x^2 - 13x + 6 = 0.$$

$$\begin{aligned} &\rightarrow \pm 1, \pm 2, \pm 3, \pm 6. \\ &\rightarrow 1, 2, 3, 6. \end{aligned}$$

$$(x-1) \cdot (x^3 - 4x^2 + 7x - 6) = 0$$

$$(x-1) \cdot (x-2) \cdot (x^2 - 2x + 3) = 0$$

$$\begin{aligned} &\Rightarrow x = \frac{2 \pm \sqrt{4-12}}{2} \\ &= 1 \pm \sqrt{2}i \end{aligned}$$

$$\alpha_1 = 1$$

$$\alpha_2 = 2$$

$$\alpha_3 = 1 + \sqrt{2}i$$

$$\alpha_4 = 1 - \sqrt{2}i$$

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

$$\hookrightarrow p \nmid q$$

$$\begin{aligned} &p \text{ divides } e \\ &q \text{ " } a. \end{aligned}$$

$$4!$$

$$\begin{array}{c} 1 \\ \boxed{2} \\ \sqrt{3} \\ \sqrt{3} \end{array}$$

$$p=2.$$

$$\boxed{3 + 2\sqrt{3}} = 2$$



Alt.

$$\underbrace{x \cdot \left( \frac{5-x}{x+1} \right)}_{=a} \cdot \underbrace{\left( x + \frac{5-x}{x+1} \right)}_{=b} = 6 \Rightarrow a \cdot b = 6.$$

$$\left. \begin{array}{l} a = \frac{5x-x^2}{x+1} \\ b = \frac{x^2+5}{x+1} \end{array} \right\} + \rightarrow a+b = \frac{5x+5}{x+1} = 5$$

$$\begin{aligned} \Rightarrow a+b &= 5 \\ x^2-5x+6 &= 0 \\ a=2 \text{ \& } b=3 & \rightarrow a, b \\ \text{OR } a=3 \text{ \& } b=2. \end{aligned}$$

C-1  $a=2 \text{ \& } b=3$

$$\begin{aligned} 5x-x^2 &= 2x+2 \text{ \& } x^2+5=3x+3 \\ x^2-3x+2 &= 0 & x^2-3x+2 &= 0 \\ \downarrow \uparrow & \\ \boxed{x=1, 2} \end{aligned}$$

C-2

$$\begin{aligned} a=3 \text{ \& } b=2. \\ 5x-x^2 &= 3x+3 \text{ \& } x^2+5=2x+2 \\ x^2-2x+3 &= 0 & x^2-2x+3 &= 0 \\ \downarrow \uparrow & \\ x &= 1 \pm \sqrt{2}i \end{aligned}$$



# Column Matching

## Column-I

(A)  $\int_1^{\infty} \frac{x^4+1}{x^2(x^2+1)^2} \cdot dx =$

(B)  $\int_0^{\infty} \left[ \frac{5}{e^x} \right] \cdot dx =$   
G.I.F.

(C)  $\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{e^x + 1} dx =$

## Column-II

(P)  $\frac{\pi}{2\sqrt{3}}$

(Q)  $\frac{6-\pi}{4}$

(R)  $\ln\left(\frac{625}{24}\right)$

(S) Can not be determined

(T) None of P, Q, R, S is correct match.

Ans

|   |   |   |
|---|---|---|
| A | → | Q |
| B | → | R |
| C | → | P |



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$$I = \int_1^{\infty} \frac{x^4 + 1}{x^2 (x^2 + 1)^2} dx = \int_1^{\infty} \frac{x^4 + 1 + 2x^2 - 2x^2}{x^2 (1 + x^2)^2} dx$$

$$= \int_1^{\infty} \frac{(x^2 + 1)^2 - 2x^2}{x^2 (1 + x^2)^2} dx = \int_1^{\infty} \frac{1}{x^2} dx - 2 \int_1^{\infty} \frac{dx}{(1 + x^2)^2}$$

$x = \tan(\theta)$   
 $dx = \sec^2(\theta) d\theta$

$$= \left[ -\frac{1}{x} \right]_1^{\infty} - 2 \int_{\pi/4}^{\pi/2} \frac{\cancel{\sec^2(\theta)} d\theta}{(\sec^2(\theta))^2}$$

$$= 0 - (-1) - \int_{\pi/4}^{\pi/2} (1 + \cos(2\theta)) d\theta = 1 - \left( \theta + \frac{\sin(2\theta)}{2} \right)_{\pi/4}^{\pi/2}$$

$$= 1 - \left( \frac{\pi}{4} - \frac{1}{2} \right) = \frac{3}{2} - \frac{\pi}{4} = \frac{6 - \pi}{4}$$



③

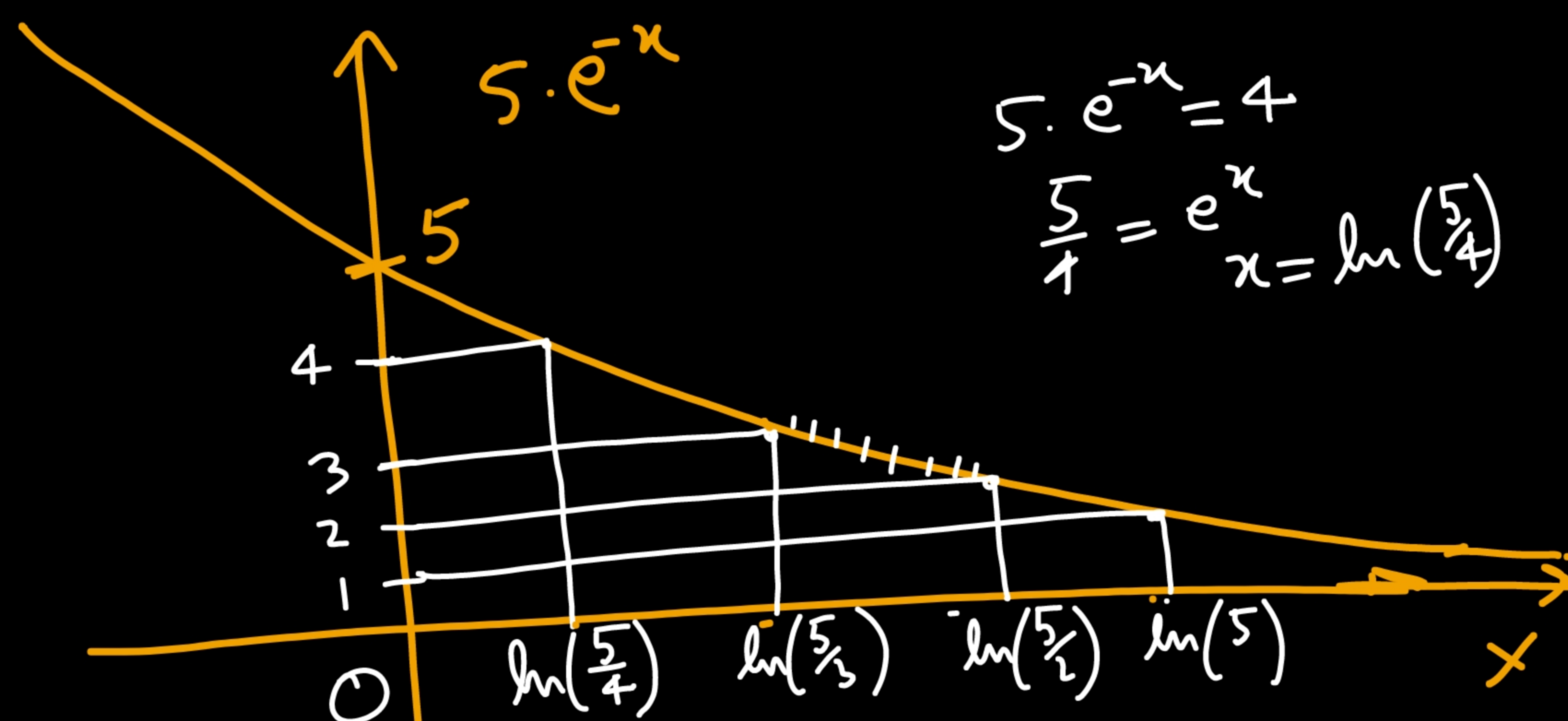
$$\int_0^{\infty} \left[ \frac{5}{e^x} \right] dx$$

$$= \int_0^{\infty} [5e^{-x}] dx$$

$$= \int_0^{\ln(\frac{5}{4})} 4 \cdot dx + \int_{\ln(\frac{5}{4})}^{\ln(\frac{5}{3})} 3 \cdot dx + \int_{\ln(\frac{5}{3})}^{\ln(\frac{5}{2})} 2 \cdot dx + \int_{\ln(\frac{5}{2})}^{\ln(5)} 1 \cdot dx + \int_{\ln(5)}^{\infty} 0 \cdot dx$$

$$= 4 \left( \ln\left(\frac{5}{4}\right) - 0 \right) + 3 \cdot \left( \ln\left(\frac{5}{3}\right) - \ln\left(\frac{5}{4}\right) \right) + 2 \left( \ln\left(\frac{5}{2}\right) - \ln\left(\frac{5}{3}\right) \right) + \ln(5) - \ln\left(\frac{5}{2}\right)$$

$$= \ln\left(\frac{5}{4}\right) + \ln\left(\frac{5}{3}\right) + \ln\left(\frac{5}{2}\right) + \ln(5) = \ln\left(\frac{5}{4} \cdot \frac{5}{3} \cdot \frac{5}{2} \cdot 5\right) = \ln\left(\frac{625}{24}\right)$$



$$5 \cdot e^{-x} = 4$$

$$\frac{5}{4} = e^x \quad x = \ln\left(\frac{5}{4}\right)$$



$$\textcircled{C} \quad \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{1+e^x} \cdot dx$$

$$\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{\frac{\pi}{2} - \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{1+e^x} dx$$

$$\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{1}{1+e^x} \cdot dx = \frac{\pi}{2} \int_0^{\frac{1}{\sqrt{3}}} \left( \frac{1}{1+e^x} + \frac{1}{1+e^{-x}} \right) dx$$

$$\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{1}{1+e^x} \cdot dx = \frac{\pi}{2} \cdot \frac{1}{\sqrt{3}} = \frac{\pi}{2\sqrt{3}}$$

Let  $x = \tan \theta \rightarrow \theta \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$   
 $2\theta \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

$$= \sin^{-1}(\sin 2\theta) = 2\theta$$

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \tan^{-1}(\tan 2\theta) = 2\theta$$

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) - \sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\theta - 2\theta = 0$$

$$\int_{-a}^a f(x) dx = \int_0^a (f(x) + f(-x)) dx$$



Ques

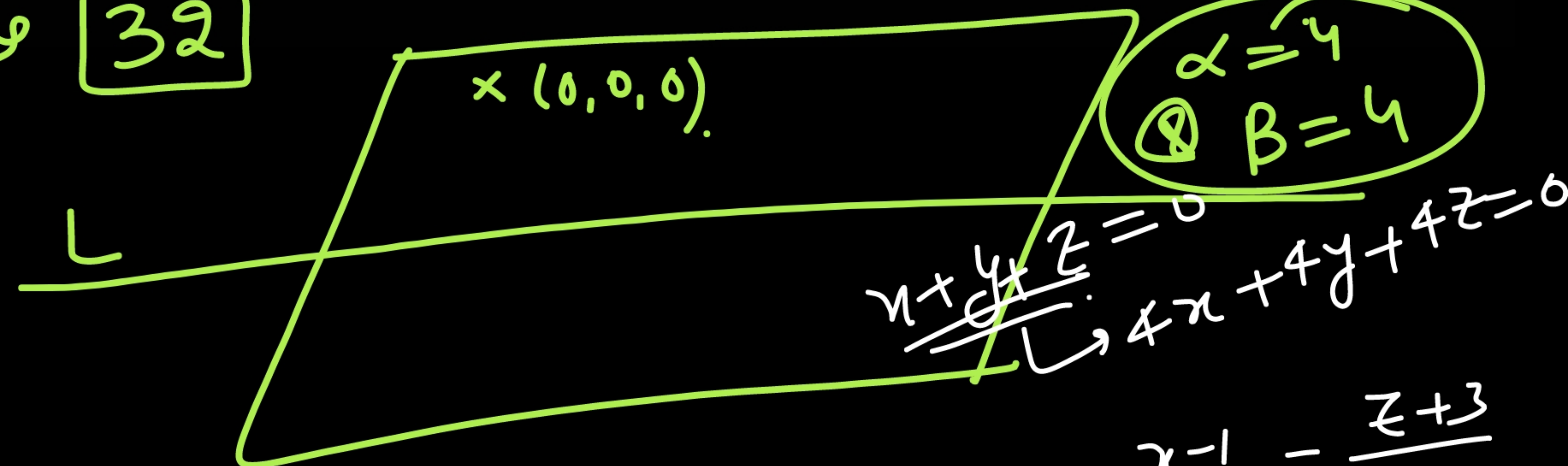
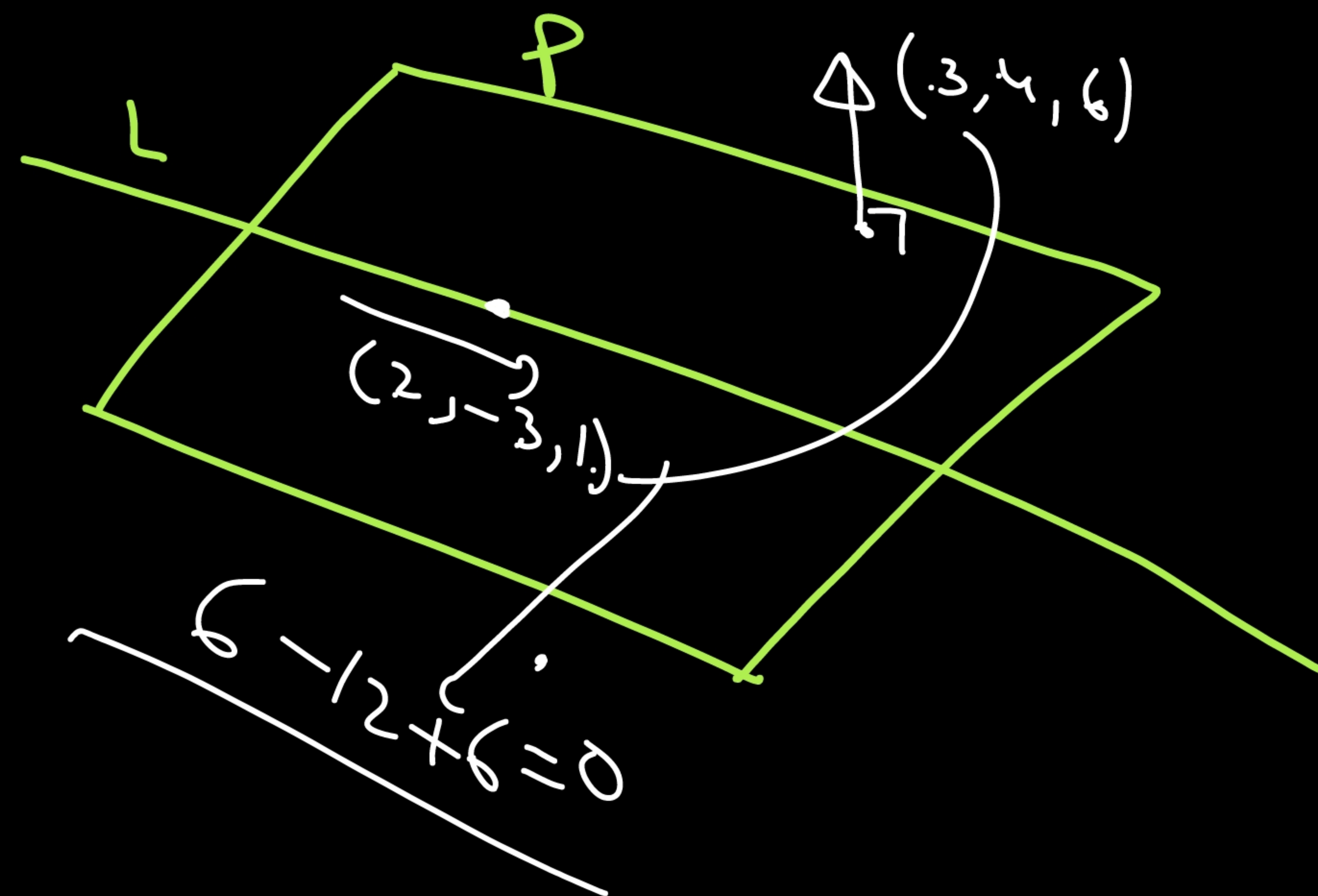
About the line  $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{1}$  the plane  $3x + 4y + 6z + 7 = 0$  is rotated till the plane passes through the origin. Now  $4x + \alpha y + \beta z = 0$  is the equation of plane in new position. The value of  $\alpha^2 + \beta^2$  is

Ans 32

$(1, 2, -3)$

$$L \equiv \frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+3}{1}$$

$$P \equiv 3x + 4y + 6z + 7 = 0$$



$$P_1 \equiv \frac{x-1}{2} = \frac{y-2}{-3}$$

$$P_2 \equiv \frac{x-1}{2} = \frac{z+3}{1}$$

$$P_3 \equiv P_1 + \lambda P_2 = 0$$

$$3(x-1) + 2(y-2) + \lambda(x-1-2(z+3)) = 0$$

$$-3-4 + \lambda(-1-6) = 0$$

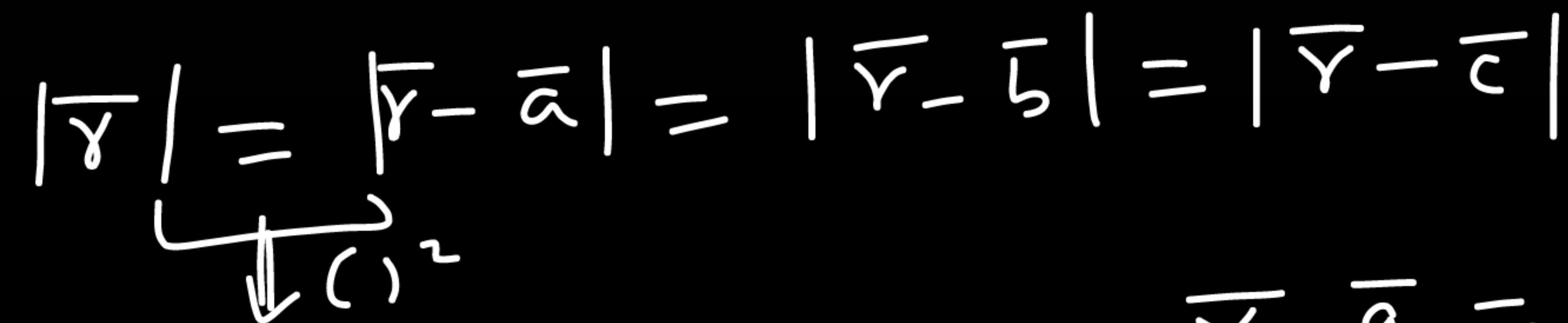
$$\lambda = -1$$

$$3x + 2y - 7 - (x - 2z - 7) = 0$$

$$2x + 2y + 2z = 0$$



If the circumcentre of the tetrahedron OABC is given by  $\frac{\vec{a}^2 (\vec{b} \times \vec{c}) + \vec{b}^2 (\vec{c} \times \vec{a}) + \vec{c}^2 (\vec{a} \times \vec{b})}{\alpha}$ , where  $\vec{a}$ ,  $\vec{b}$  &  $\vec{c}$  are the position vectors of the points A, B, C respectively relative to the origin 'O' such that  $[\vec{a} \ \vec{b} \ \vec{c}] = 36$  then  $\alpha$  is



Let  $\bar{Y} = \lambda_1 \bar{a} + \lambda_2 \bar{b} + \lambda_3 \bar{c} = \mu_1 \bar{a} \times \bar{b} + \mu_2 \bar{b} \times \bar{c} + \mu_3 \bar{c} \times \bar{a}$

$$\vec{r} = \mu_1 \vec{a} \times \vec{b} + \mu_2 \vec{b} \times \vec{c} + \mu_3 \vec{c} \times \vec{a} \xrightarrow{\cdot \vec{c}} \vec{r} \cdot \vec{c} = \mu_1 [\vec{a} \vec{b} \vec{c}]$$

$$\mu_2 = \frac{\sigma^2}{2[\bar{a}\bar{b}\bar{c}]}$$

$$\textcircled{8} \quad \mu_3 = \frac{b^2}{2[\bar{a} \bar{b} \bar{c}]}$$

$$\frac{C^2}{2} = M_1 \left( \frac{a}{b} \right)$$

$$\mu_k = \frac{c^k}{2[\bar{a} \quad \bar{b} \quad \bar{c}]}$$



$$\overline{\gamma} = \frac{a^2 \overline{b} \times \overline{c} + b^2 \overline{c} \times \overline{a} + c^2 \overline{a} \times \overline{b}}{2[\overline{a} \overline{b} \overline{c}]}$$

Comparing

$$\alpha = 2[\overline{a} \overline{b} \overline{c}] = 2 \times 36 = \boxed{72}$$



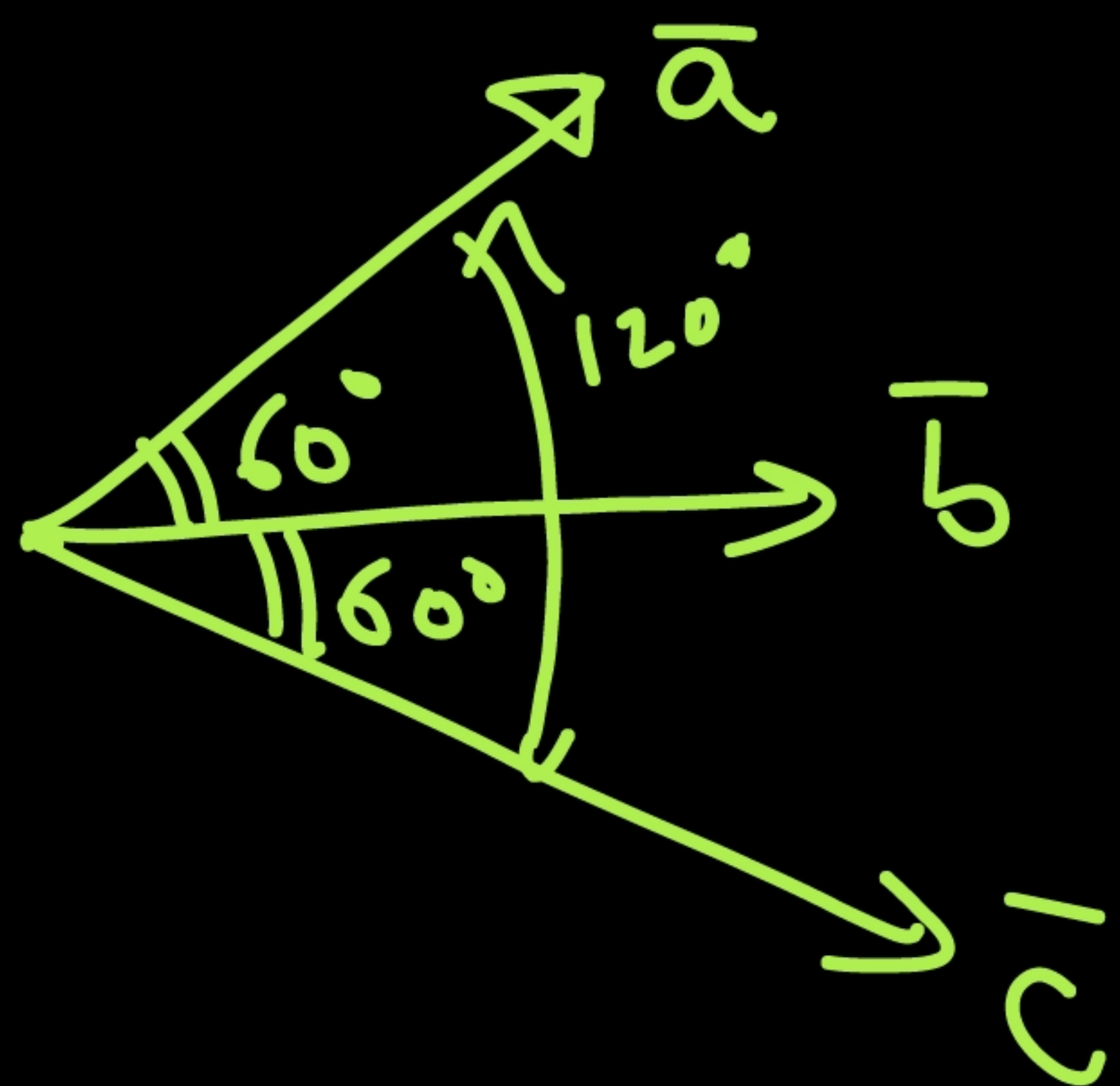
Ques

Given four non zero vectors  $\vec{a}$ ,  $\vec{b}$ ,  $\vec{c}$  and  $\vec{d}$ . The vectors  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  are coplanar but not collinear pair by pair and vector  $\vec{d}$  is not coplanar with vectors  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  and  $(\vec{a} \wedge \vec{b}) = (\vec{b} \wedge \vec{c}) = \frac{\pi}{3}$ ,  $(\vec{d} \wedge \vec{a}) = \alpha$

and  $(\vec{d} \wedge \vec{b}) = \beta$ , if  $(\vec{d} \wedge \vec{c}) = \cos^{-1}(m \cos \beta + n \cos \alpha)$  then  $m - n$  is :

2

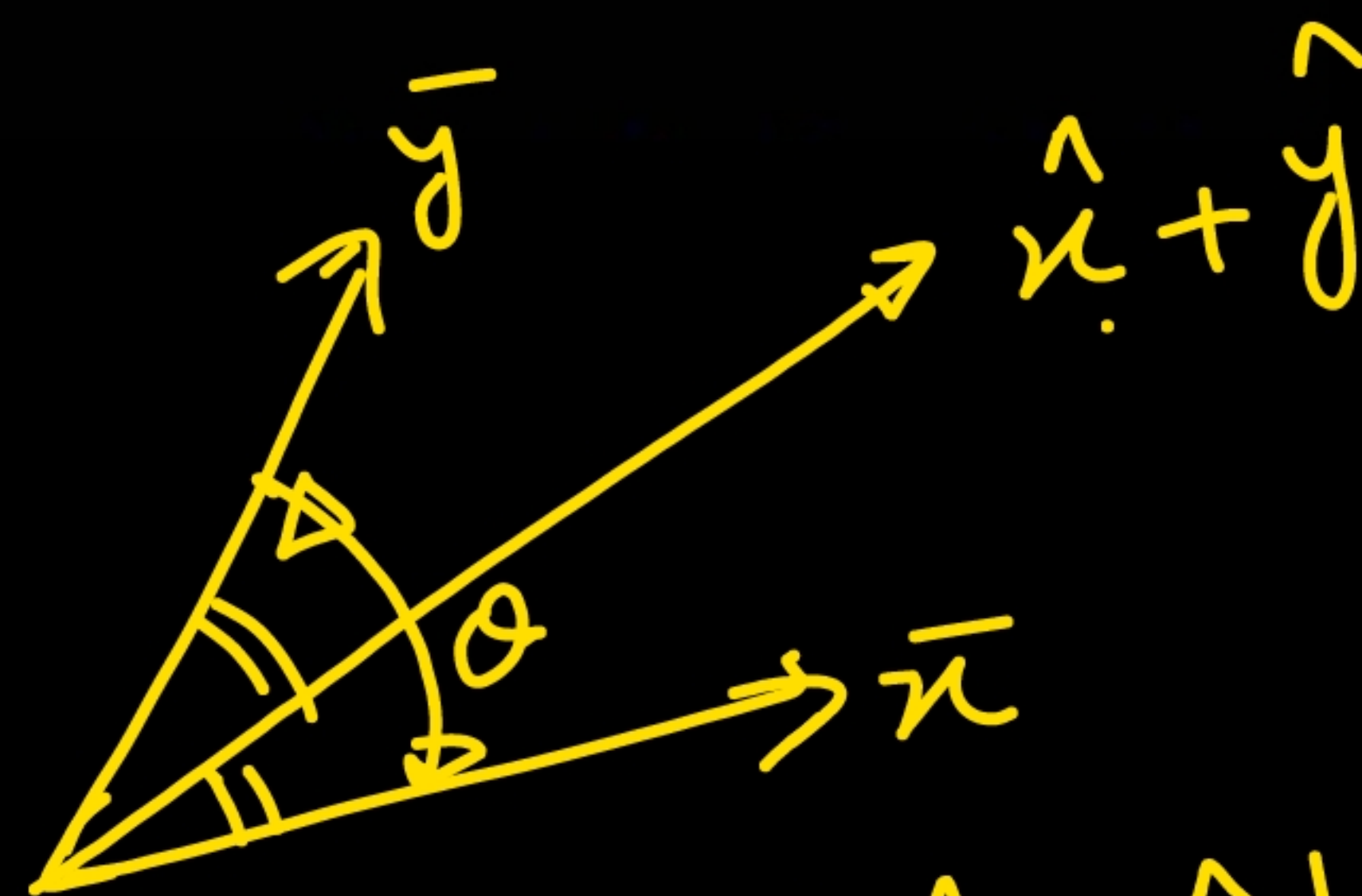
$$(\vec{a}, \vec{b}) = (\vec{b}, \vec{c}) = \frac{\pi}{3}$$



$$\vec{b} = \frac{\vec{a} + \vec{c}}{2 \cos\left(\frac{120^\circ}{2}\right)} \Rightarrow \vec{b} = \vec{a} + \vec{c} \Rightarrow \vec{c} = \vec{b} - \vec{a}$$

$$(\vec{d} \wedge \vec{c}) = \theta = \cos^{-1}(\cos \beta - \cos \alpha)$$

$$m=1, n=-1 \Rightarrow m-n=2$$



$$|\vec{x} + \vec{y}| = 2 \cos\left(\frac{\theta}{2}\right)$$

$$\vec{c} \cdot \vec{d} = \vec{b} \cdot \vec{d} - \vec{a} \cdot \vec{d}$$

$$\cos(\theta) = \cos(\beta) - \cos(\alpha)$$



Ques

Let  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  be non-coplanar unit vectors equally inclined to one another at an acute angle  $\theta$ . Then  $|\vec{a} \ \vec{b} \ \vec{c}|$  in terms of  $\theta$  is equal to:

(A)  $(1 + \cos \theta) \sqrt{\cos 2\theta}$

(B)  $(1 + \cos \theta) \sqrt{1 - 2\cos 2\theta}$

✓ (C\*)  $(1 - \cos \theta) \sqrt{1 + 2\cos \theta}$

(D)  $(1 - \sin \theta) \sqrt{1 + 2\cos \theta}$

Sol

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos(\theta)$$

As we know

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{a} \cdot \vec{c} & \vec{b} \cdot \vec{c} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{vmatrix}$$

$$= 1 + 2\cos^3 \theta - 3\cos^2 \theta = (1 - \cos \theta)(1 + \cos \theta - 2\cos^2 \theta)$$

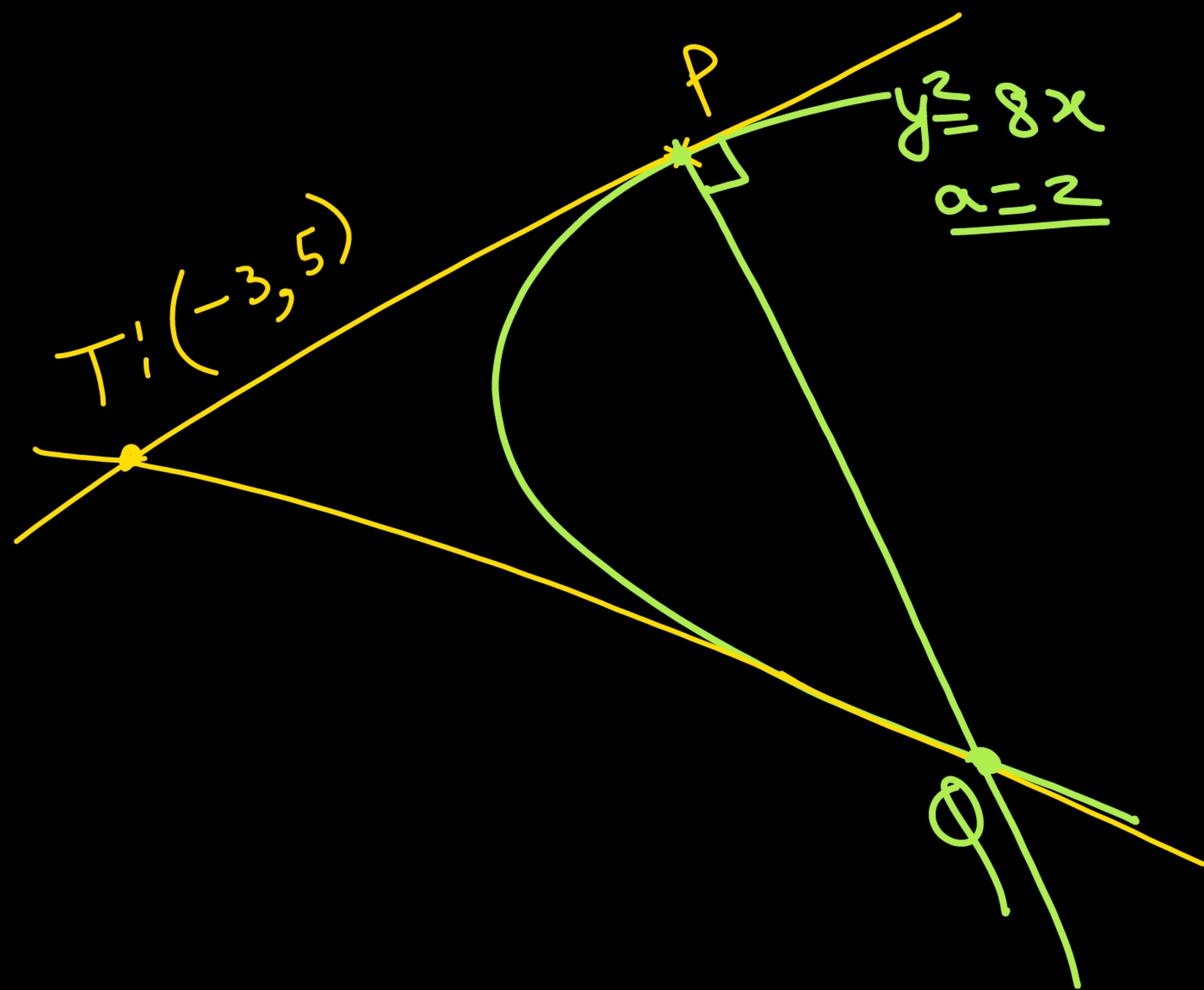
$$= (1 - \cos \theta)(1 - \cos \theta)(1 + 2\cos \theta)$$

$$= (1 - \cos \theta)^2 (1 + 2\cos \theta) \Rightarrow |\vec{a} \ \vec{b} \ \vec{c}| = (1 - \cos \theta) \sqrt{1 + 2\cos \theta}$$



**Ques** Let  $P, Q$  are the points of contact of tangents drawn from a point  $T(-3, 5)$  to the parabola  $y^2 = 8x$ . If  $PQ$  is normal to the parabola at  $P$ , then magnitude of  $x$ -coordinate of mid point of  $TP$  is

(A) 0      (B) 1      ☒ (C) 2      (D) None of these.



$$\begin{aligned} \text{Let } P \text{ is } (at_1^2, 2at_1) \\ Q \text{ is } (at_2^2, 2at_2) \end{aligned} \Rightarrow t_2 = -t_1 - \frac{2}{t_1}$$

$$\Rightarrow t_1 t_2 + t_1^2 = -2$$

$$T \equiv (at_1 t_2, a(t_1 + t_2))$$

$$\text{Mid pt of } TP \equiv \left( \frac{at_1 t_2 + at_1^2}{2}, \frac{a(t_1 + t_2 + t_1)}{2} \right) \equiv M$$

$$\begin{aligned} \text{Required} = |x_M| &= \left| \frac{at_1 t_2 + at_1^2}{2} \right| = |t_1 t_2 + t_1^2| \\ &= |-2| = \boxed{2} \end{aligned}$$



Ques

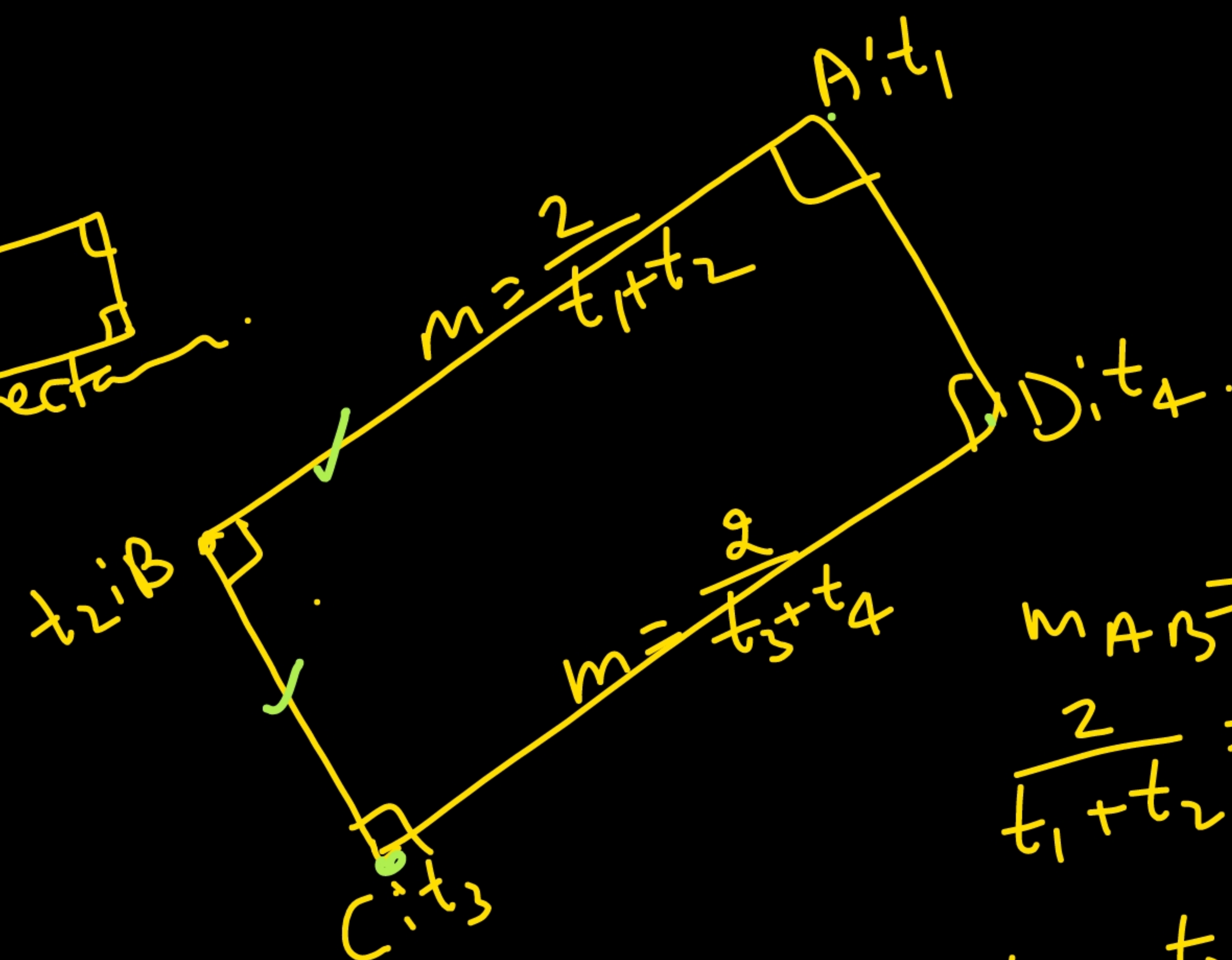
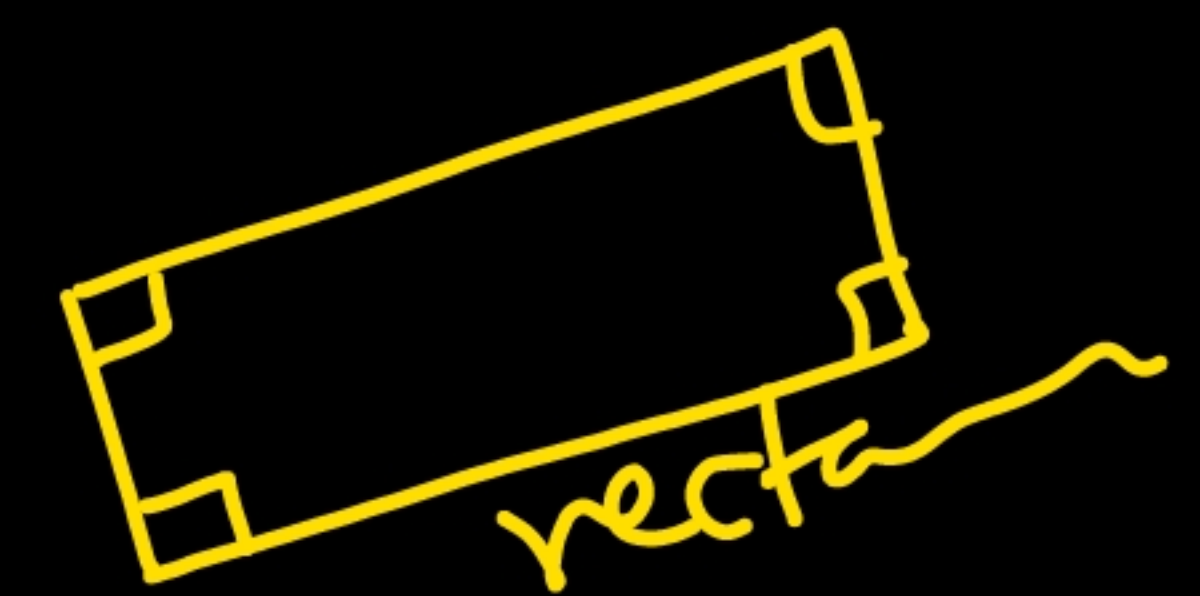
A quadrilateral is inscribed in a parabola, then

(A) quadrilateral may be cyclic

(B) diagonals of the quadrilateral may be equal

(C) all possible pairs of adjacent sides may be perpendicular

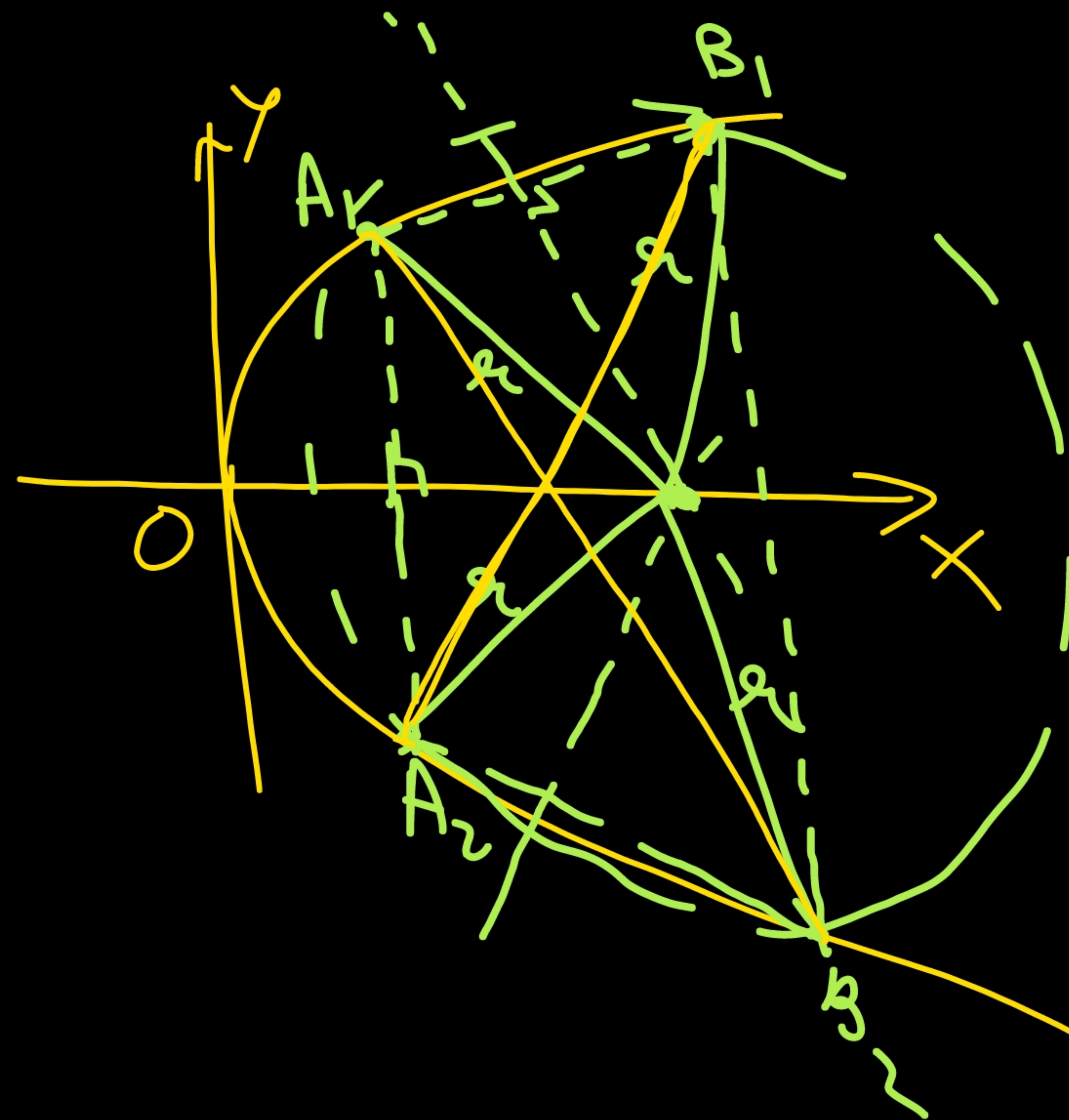
(D) none of these.



$$m_{AB} = m_{CD}$$

$$\frac{2}{t_1+t_2} = \frac{2}{t_3+t_4}$$

$$t_1+t_2 = t_3+t_4 \quad \text{--- (1)}$$



$$m_{BC} \cdot m_{AD} = -1$$

$$\frac{2}{t_1+t_2} \cdot \frac{2}{t_2+t_3} = -1$$

$$t_2+t_3 = -\frac{4}{t_1+t_2} \quad \parallel \text{ by } y$$

$$t_3 = -t_2 - \frac{4}{t_1+t_2}$$

$$t_4 = t_1 - \frac{4}{t_1+t_2}$$



$$t_3 = -t_2 - \frac{4}{t_1+t_2}$$

$$t_4 = -t_1 - \frac{4}{t_1+t_2}$$

$$\begin{array}{r} + \\ \hline t_3+t_4 = -(t_1+t_2) - \frac{8}{t_1+t_2} \end{array}$$

$$t_3+t_4 = - \downarrow (t_1+t_2)$$

$$2(t_1+t_2) = -\frac{8}{t_1+t_2}$$

$$(t_1+t_2)^2 = -4 \quad \times$$